

Greedy Templates

Two interval templates plus non-interval greedy patterns.

Quick Reference Table

#	Name	Description	Intuition	Time	Space	Key Implementation
435	Non-overlapping Intervals	Remove the minimum number of intervals to make the rest non-overlapping.	Always keep the interval that finishes earliest — it leaves the most room for the rest, so greedily picking by end maximizes how many you keep.	$O(n \log n)$	$O(1)$	Standard
452	Minimum Number of Arrows to Burst Balloons	Balloons span $[start, end]$. An arrow shot at x pops all balloons with $start \leq x \leq end$. Minimum arrows to pop all balloons.	Shoot the arrow at the earliest end point to pop every balloon overlapping it; only start a new arrow when a balloon begins past the current arrow.	$O(n \log n)$	$O(1)$	Standard
646	Maximum Length of Pair Chain	Given pairs $[a, b]$, form a chain where $b < c$ for each consecutive pair $[a, b] \rightarrow [c, d]$. Maximum chain length.	Picking the pair that ends earliest each time leaves the most slack for later links, maximizing the chain.	$O(n \log n)$	$O(1)$	Standard
56	Merge Intervals	Merge all overlapping intervals. Return the minimum set of non-overlapping intervals.	sort to make overlaps adjacent, then sweep once — each interval only ever compares against the most recently kept one.	$O(n \log n)$	$O(n)$	Standard
57	Insert Interval	Insert a new interval into a sorted list of non-overlapping intervals; merge if necessary. The input is already sorted — no sort needed.	Walk the sorted list in three phases — copy everything that ends before the new interval, absorb everything that overlaps it, then copy the rest.	$O(n)$	$O(n)$	Standard
253	Meeting Rooms II	Minimum number of meeting rooms required to host all meetings.	USE WHEN counting simultaneous/overlapping resources — "minimum rooms", "max concurrent". The heap size at any moment is exactly how many meetings are running at once.	$O(n \log n)$	$O(n)$	Standard
55	Jump Game	Each element is the max jump from that position. Can you reach the last index?	Sweep left to right tracking the farthest index reachable; if you ever stand past that frontier you're stuck.	$O(n)$	$O(1)$	Standard
45	Jump Game II	Minimum jumps to reach the last index.	Treat each jump as a BFS layer — extend through the whole current reach before incrementing the jump count.	$O(n)$	$O(1)$	Standard
763	Partition Labels	Partition string into as many parts as possible so each letter appears in at most one part. Return the size of each part.	A partition can't close until you pass the last occurrence of every letter seen inside it, so extend <code>end</code> to that frontier and cut when you reach it.	$O(n)$	$O(1)$	Standard
406	Queue Reconstruction by Height	People given as $[h, k]$ (height h , k taller/equal people in front). Reconstruct the queue.	Place tallest first so that inserting a person at index <code>k</code> is always correct — only taller/equal people are already placed, and shorter insertions	$O(n^2)$	$O(n)$	Standard

#	Name	Description	Intuition	Time	Space	Key Implementation
			later don't disturb their counts.			
252	Meeting Rooms	Given an array of meeting intervals, determine if a person could attend all meetings (no two meetings overlap).	After sorting by start, any overlap must be between adjacent intervals, so one linear scan comparing neighbors is enough.	$O(n \log n)$	$O(1)$	start < previous end = overlap
134	Gas Station	There are n gas stations on a circular route. <code>gas[i]</code> is the gas at station i ; <code>cost[i]</code> is the gas needed to travel from i to $i+1$. Return the starting station index if you can complete the circuit, or -1 if not.	If the tank ever drops below 0, no station between the old start and here can work, so jump the start past the failure point; a valid start exists iff total gas $\geq$ total cost.	$O(n)$	$O(1)$	reset start when tank goes negative
135	Candy	Assign minimum candies to children in a line: each child gets at least 1 candy, and a child with a higher rating than an adjacent neighbor gets more candies than that neighbor.	Two passes capture the two directions of the constraint — a left-to-right pass enforces "higher than left neighbor gets more", a right-to-left pass enforces "higher than right neighbor gets more". Taking the max of both satisfies both.	$O(n)$	$O(n)$	higher than left neighbor
274	H-Index	Given an array of citation counts, find the largest h such that at least h papers each have at least h citations.	After sorting ascending, if a paper at index i has <code>citations[i]</code> citations, then there are $n - i$ papers with at least that many. The h-index is found where <code>citations[i] $\geq$ $n - i$</code> first holds.	$O(n \log n)$	$O(1)$	$n-i$ papers have $\geq$ citations[i]
455	Assign Cookies	Each child i has a greed factor <code>g[i]</code> (minimum cookie size needed); each cookie j has size <code>s[j]</code> . Each cookie satisfies at most one child. Maximize the number of content children.	Sort both arrays; give the smallest sufficient cookie to the least greedy child. This never wastes a large cookie on a child a small one could satisfy.	$O(n \log n)$	$O(1)$	Standard
575	Distribute Candies	Distribute n candies (the holder eats exactly $n/2$) to maximize the number of distinct candy types eaten.	You can eat at most $n/2$ candies, and at best one of each distinct type. So the answer is the smaller of the number of distinct types and $n/2$.	$O(n)$	$O(n)$	capped by half the total
605	Can Place Flowers	Given a flowerbed (array of 0s and 1s where 1 is a planted flower), determine if n new flowers can be planted without any two flowers being adjacent.	Greedily plant a flower at the first empty plot whose neighbors are also empty — planting as early as possible never blocks a future placement that a later choice would allow.	$O(n)$	$O(1)$	plot and both neighbors empty
630	Course Schedule III	Each course has (<code>duration</code> , <code>lastDay</code>). Starting at day 0 and taking courses	Sort by deadline so deadlines are considered in order. Greedily take each course; if total time exceeds the current	$O(n \log n)$	$O(n)$	over deadline $\rightarrow$ drop longest

#	Name	Description	Intuition	Time	Space	Key Implementation
		one at a time, find the maximum number of courses that can be taken so each finishes by its <code>lastDay</code> .	deadline, drop the previously taken course with the largest duration (a max-heap) — swapping out the longest course frees the most time while keeping the count.			
881	Boats to Save People	Each boat carries at most 2 people with total weight at most <code>limit</code> . Given people's weights, find the minimum number of boats to rescue everyone.	Pair the heaviest remaining person with the lightest remaining one if they fit together; otherwise the heaviest goes alone. Sorting plus two pointers makes this pairing optimal.	$O(n \log n)$	$O(1)$	Standard
986	Interval List Intersections	Given two lists of pairwise-disjoint, sorted intervals, return the intersection of the two lists.	Two pointers walk both sorted lists together. The overlap of the two current intervals is <code>[max(starts), min(ends)]</code> , which is valid only when <code>max(starts) <= min(ends)</code> . Advance whichever interval ends first.	$O(m + n)$	$O(m + n)$	valid overlap
1005	Maximize Sum Of Array After K Negations	Given an array and <code>k</code> , perform exactly <code>k</code> operations, each negating one element (the same index may be chosen repeatedly), to maximize the array sum.	Flipping the most negative number first yields the biggest gain. After all negatives are flipped (or <code>k</code> runs out), any leftover flips should land on the smallest-magnitude element, since flipping it twice is a no-op.	$O(n \log n)$	$O(1)$	leftover odd flips hit smallest
1094	Car Pooling	Given trips <code>[numPassengers, from, to]</code> and a car capacity, determine whether all trips can be completed without exceeding capacity at any point.	A difference array (sweep line) over locations records passengers boarding at <code>from</code> and leaving at <code>to</code> . Running the prefix sum gives the occupancy at each point; if it ever exceeds capacity, fail.	$O(n + \text{maxLocation})$	$O(\text{maxLocation})$	over capacity → fail
1272	Remove Interval	Given a sorted list of disjoint intervals and an interval <code>toBeRemoved</code> , return the set of intervals after removing every point that lies inside <code>toBeRemoved</code> .	Each interval is either fully outside the removal range (keep as is) or partially overlaps it (keep the non-overlapping left and/or right slivers). Walk once and emit the surviving pieces.	$O(n)$	$O(n)$	keep left sliver
1326	Minimum Number of Taps to Open to Water a Garden	A garden spans <code>[0, n]</code> . Tap <code>i</code> at position <code>i</code> waters <code>[i - ranges[i], i + ranges[i]]</code> . Find the minimum number of taps to water the whole garden, or -1 if impossible.	This is a jump-game in disguise: for each starting point record the farthest reach of any tap covering it, then greedily extend coverage layer by layer like minimum jumps.	$O(n)$	$O(n)$	gap in coverage → impossible
1353	Maximum Number of Events That Can Be Attended	Each event <code>[start, end]</code> can be attended on any single day in <code>[start, end]</code> . Attend at most one event per day.	Process days in increasing order; among all events available on the current day, attend the one that ends soonest (a min-heap by end	$O(n \log n)$	$O(n)$	attend earliest-ending event

#	Name	Description	Intuition	Time	Space	Key Implementation
		Find the maximum number of events you can attend.	day) — saving longer-lived events for later days.			
1488	Avoid Flood in The City	<code>rains[i] > 0</code> means lake <code>rains[i]</code> fills with rain on day <code>i</code> ; <code>rains[i] == 0</code> is a dry day on which you may dry exactly one lake. A full lake that rains again floods. Return an array where dry days hold the lake to dry (any valid choice) and rain days hold -1, or an empty array if flooding is unavoidable.	When a lake that is already full rains again, you must have used a dry day between its two rains. Greedily assign the earliest available dry day that falls after the lake was last filled — a <code>TreeSet</code> of dry-day indices gives that earliest day.	$O(n \log n)$	$O(n)$	no dry day available → flood

Two Interval Templates

MERGE DIRECTION RULE (for merge-style problems like #56, where both keys work):

Sort by START → traverse FORWARD ($i = 0 \rightarrow n-1$), extend the END of last kept

Sort by END → traverse BACKWARD ($i = n-1 \rightarrow 0$), extend the START of last kept

These two are exact mirror images. See #56 for both written out side by side.

(NOTE: this mirror only applies to merging. Template 1 below sorts by end but still sweeps FORWARD — there the goal is counting, not merging.)

TEMPLATE 1 — Sort by END time, sweep forward, track end of last kept interval

Use when: maximizing count of non-overlapping intervals

Greedy insight: earliest-finishing interval leaves most room for future ones

TEMPLATE 2 — Sort by START time, sweep forward, merge when overlapping

Use when: merging/counting overlapping intervals

Greedy insight: processing in start order → each interval only needs to compare with the last merged result

Variables: `intervals` = list of `[start, end]` · `lastEnd` = end of last kept interval · `result` = merged output · `pq` = min-heap of active end times **Pseudocode:**

TEMPLATE 1 (sort by end, count non-overlapping):

```
sort intervals by END time
lastEnd = -infinity
for each interval:
    if interval.start >= lastEnd:    # no overlap
        lastEnd = interval.end
        keep it (count++ or add)
    else:
        skip it (overlap)
```

TEMPLATE 2 (sort by start, merge):

```
sort intervals by START time
for each interval:
    if result empty OR last kept end < interval.start:
        add interval as new
    else:
        extend last kept end = max(last end, interval.end)
```

TEMPLATE 3 (sort by start + min-heap of ends):

```
sort intervals by START time
for each interval:
    if heap nonempty AND earliest end <= interval.start:
        pop heap                # reuse freed resource
    push interval.end           # occupy a resource
heap size = resources needed
```

```

// sort to expose a local greedy choice, then sweep once making the locally-best pick. - WHEN: interval
// TEMPLATE 1: Sort by end, compare start
Arrays.sort(intervals, (a, b) -> a[1] - b[1]); // sort by END time
int lastEnd = Integer.MIN_VALUE;
for (int[] interval : intervals) {
    if (interval[0] >= lastEnd) { // start >= lastEnd → no overlap → keep
        lastEnd = interval[1];
        // count++ or add to result
    } else {
        // overlap → skip (or count as removed)
    }
}

// TEMPLATE 2: Sort by start, compare end
Arrays.sort(intervals, (a, b) -> a[0] - b[0]); // sort by START time
List<int[]> result = new ArrayList<>();
for (int[] interval : intervals) {
    if (result.isEmpty() || result.get(result.size()-1)[1] < interval[0]) {
        result.add(interval); // no overlap → new interval
    } else {
        result.get(result.size()-1)[1] =
            Math.max(result.get(result.size()-1)[1], interval[1]); // overlap → extend end
    }
}

// TEMPLATE 3: Sort by start + min-heap of end times (multi-resource)
// USE WHEN counting simultaneous/overlapping resources - "minimum rooms", "max concurrent".
Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
PriorityQueue<Integer> pq = new PriorityQueue<>(); // min-heap of active end times
for (int[] interval : intervals) {
    if (!pq.isEmpty() && pq.peek() <= interval[0])
        pq.poll(); // reuse: earliest-ending resource is free
    pq.offer(interval[1]); // occupy a resource until interval[1]
}
// pq.size() = resources needed

```

Part 1 — Sort by End Time

#435 Non-overlapping Intervals

Description: Remove the minimum number of intervals to make the rest non-overlapping.

Intuition: Always keep the interval that finishes earliest — it leaves the most room for the rest, so greedily picking by end maximizes how many you keep.

Key: equivalent to keeping the maximum number of non-overlapping intervals. Sort by end, greedily keep each interval whose start $\geq$ last kept end.

Variables: `intervals` = [start, end] pairs · `kept` = count of non-overlapping intervals kept · `lastEnd` = end of last kept interval **Pseudocode:**

```

sort intervals by END
kept = 0, lastEnd = -infinity
for each interval:
    if interval.start >= lastEnd:    # no overlap with last kept
        lastEnd = interval.end
        kept++
    # else overlap -> skip (it is removed)
return total intervals - kept

```

```

class Solution {
    public int eraseOverlapIntervals(int[][] intervals) {
        Arrays.sort(intervals, (a, b) -> a[1] - b[1]);    // sort by END
        int kept = 0, lastEnd = Integer.MIN_VALUE;
        for (int[] interval : intervals) {
            if (interval[0] >= lastEnd) {    // no overlap -> keep
                lastEnd = interval[1];
                kept++;
            }
            // else: overlap -> skip (counted as removed)
        }
        return intervals.length - kept;
    }
}

```

Time $O(n \log n)$ | **Space** $O(1)$

#452 Minimum Number of Arrows to Burst Balloons

Description: Balloons span $[start, end]$. An arrow shot at x pops all balloons with $start \leq x \leq end$. Minimum arrows to pop all balloons.

Intuition: Shoot the arrow at the earliest end point to pop every balloon overlapping it; only start a new arrow when a balloon begins past the current arrow.

Key: same structure as #435 — sort by end, count groups. One arrow suffices per group. New group starts when $start > \text{current arrow position (= last group's end)}$.

Variables: `points` = balloon $[start, end]$ ranges · `arrows` = arrows used so far · `arrowEnd` = position (end) of current arrow
Pseudocode:

```

sort points by END
arrows = 1, arrowEnd = first balloon's end
for each point:
    if point.start > arrowEnd:    # this balloon not popped by current arrow
        arrows++
        arrowEnd = point.end    # new arrow at this balloon's end
    # else same arrow covers it
return arrows

```



```

class Solution {
    public int findMinArrowShots(int[][] points) {
        Arrays.sort(points, (a, b) -> Integer.compare(a[1], b[1])); // sort by END (use compare to avoid overflow)
        int arrows = 1, arrowEnd = points[0][1];
        for (int[] point : points) {
            if (point[0] > arrowEnd) { // new group: no overlap with current
                arrows++;
                arrowEnd = point[1];
            }
            // else: same arrow covers this balloon (start <= arrowEnd)
        }
        return arrows;
    }
}

```

Time $O(n \log n)$ | **Space** $O(1)$

#646 Maximum Length of Pair Chain

Description: Given pairs $[a, b]$, form a chain where $b < c$ for each consecutive pair $[a, b] \rightarrow [c, d]$. Maximum chain length.

Intuition: Picking the pair that ends earliest each time leaves the most slack for later links, maximizing the chain.

Key: identical to #435 / #452 — sort by end, greedily pick non-overlapping pairs.

Variables: $\text{pairs} = [a, b]$ pairs · $\text{count} = \text{chain length}$ · $\text{lastEnd} = \text{end (b) of last chosen pair}$ **Pseudocode:**

```

sort pairs by END (b)
count = 0, lastEnd = -infinity
for each pair:
    if pair.start > lastEnd: # strict: chain needs b < c
        count++
        lastEnd = pair.end
return count

```

```

class Solution {
    public int findLongestChain(int[][] pairs) {
        Arrays.sort(pairs, (a, b) -> a[1] - b[1]); // sort by END
        int count = 0, lastEnd = Integer.MIN_VALUE;
        for (int[] pair : pairs) {
            if (pair[0] > lastEnd) { // strict: pair[0] > lastEnd
                count++;
                lastEnd = pair[1];
            }
        }
        return count;
    }
}

```

Time $O(n \log n)$ | **Space** $O(1)$

435 vs 452 vs 646 — The One Difference

	#435 Non-overlapping	#452 Min Arrows	#646 Pair Chain
Sort by:	end	end	end
Keep condition:	start >= lastEnd	start > arrowEnd	start > lastEnd
Boundary:	>= (inclusive)	> (exclusive)	> (exclusive)
Return:	intervals.length - kept arrows (groups)		count (groups)

Why 435 uses >= but 452/646 use >:

- #435: touching intervals [1,2], [2,3] are non-overlapping (gap = 0 is OK)
- #452: arrow at x=2 pops BOTH [1,2] and [2,3] (touching = same arrow)
- #646: pair chain requires b < c strictly (touching b==c not allowed)

Part 2 — Sort by Start Time

#56 Merge Intervals

Description: Merge all overlapping intervals. Return the minimum set of non-overlapping intervals.

Intuition: sort to make overlaps adjacent, then sweep once — each interval only ever compares against the most recently kept one.

The direction rule — sort key dictates traversal direction:

- **Sort by START** → **traverse forward** ($i = 0 \rightarrow n-1$). The newest interval can only extend the last kept interval's **END**, so you compare `interval.start` against `last.end` and update `last[1]`.
- **Sort by END** → **traverse backward** ($i = n-1 \rightarrow 0$). The newest interval can only extend the last kept interval's **START**, so you compare `interval.end` against `last.start` and update `last[0]`.

Both are symmetric and equally valid. Pick the one whose sort key you already have.

Method A — Sort by START, traverse forward

Compare current `start` to last kept `end`; on overlap, extend the **end**.

Variables: `intervals` = [start, end] pairs · `result` = merged intervals · `last` = most recently kept interval

Pseudocode:

```
sort intervals by START
for each interval (forward):
  if result empty:
    add interval; continue
  last = last interval in result
  if last.end < interval.start:  # no overlap
    add interval
  else:
    last.end = max(last.end, interval.end)  # overlap -> extend END
return result
```

```

class Solution {
    public int[][] merge(int[][] intervals) {
        Arrays.sort(intervals, (a, b) -> a[0] - b[0]); // sort by START
        List<int[]> result = new ArrayList<>();
        for (int[] interval : intervals) { // ← traverse FORWARD
            if (result.isEmpty()) {
                result.add(interval);
                continue;
            }
            int[] last = result.get(result.size() - 1);
            if (last[1] < interval[0]) { // no overlap (last.end < current.start)
                result.add(interval);
            } else {
                last[1] = Math.max(last[1], interval[1]); // overlap → extend END
            }
        }
        return result.toArray(new int[0][]);
    }
}

```

Method B — Sort by END, traverse backward

Mirror image: compare current `end` to last kept `start`; on overlap, extend the **start**.

Variables: `intervals` = `[start, end]` pairs · `result` = merged intervals · `last` = most recently kept interval ·
`current` = interval at index `i` **Pseudocode:**

```

sort intervals by END
for i from n-1 down to 0 (backward):
    if result empty:
        add intervals[i]; continue
    last = last interval in result
    current = intervals[i]
    if last.start > current.end: # no overlap
        add current
    else:
        last.start = min(last.start, current.start) # overlap → extend START
return result

```

```

class Solution {
    public int[][] merge(int[][] intervals) {
        Arrays.sort(intervals, (a, b) -> a[1] - b[1]); // sort by END
        List<int[]> result = new ArrayList<>();
        for (int i = intervals.length - 1; i >= 0; i--) { // ← traverse BACKWARD
            if (result.isEmpty()) {
                result.add(intervals[i]);
                continue;
            }
            int[] last = result.get(result.size() - 1);
            int[] current = intervals[i];
            if (last[0] > current[1]) { // no overlap (last.start > current.end)
                result.add(current);
            } else {
                last[0] = Math.min(last[0], current[0]); // overlap → extend START
            }
        }
        return result.toArray(new int[0][]);
    }
}

```

Time $O(n \log n)$ | **Space** $O(n)$

#57 Insert Interval

Description: Insert a new interval into a sorted list of non-overlapping intervals; merge if necessary. The input is already sorted — no sort needed.

Intuition: Walk the sorted list in three phases — copy everything that ends before the new interval, absorb everything that overlaps it, then copy the rest.

Key: three phases: (1) add all intervals that end before new starts, (2) merge all overlapping, (3) add remaining.

Variables: `intervals` = sorted disjoint `[start, end]` · `newInterval` = interval to insert (grows as it absorbs overlaps) · `result` = output list **Pseudocode:**

```
for each interval:
    if interval.end < newInterval.start:    # entirely before new
        add interval as is
    else if interval.start > newInterval.end: # entirely after new
        flush newInterval; then newInterval = interval
    else: # overlap -> absorb into newInterval
        newInterval.start = min(both starts)
        newInterval.end = max(both ends)
add newInterval (the last pending one)
return result
```

```
class Solution {
    public int[][] insert(int[][] intervals, int[] newInterval) {
        List<int[]> result = new ArrayList<>();
        for (int[] interval : intervals) {
            if (interval[1] < newInterval[0]) {           // ends before new → keep as is
                result.add(interval);
            } else if (interval[0] > newInterval[1]) {    // starts after new → flush new, move
                result.add(newInterval);
                newInterval = interval;
            } else {                                     // overlap → absorb into new
                newInterval[0] = Math.min(newInterval[0], interval[0]);
                newInterval[1] = Math.max(newInterval[1], interval[1]);
            }
        }
        result.add(newInterval);
        return result.toArray(new int[result.size()][2]);
    }
}
```

Time $O(n)$ | **Space** $O(n)$

Part 3 — Sort by Start + Min-Heap (Multi-Resource)

#253 Meeting Rooms II

Description: Minimum number of meeting rooms required to host all meetings.

Intuition: USE WHEN counting simultaneous/overlapping resources — "minimum rooms", "max concurrent". The heap size at any moment is exactly how many meetings are running at once.

Template: sort by start. Min-heap tracks end times of active meetings. When a new meeting starts after the earliest-ending meeting finishes, reuse that room.

Variables: `intervals` = meeting `[start, end]` · `pq` = min-heap of end times of meetings currently using a room

Pseudocode:

```
sort intervals by START
pq = empty min-heap of end times
for each interval:
    if pq nonempty AND earliest end <= interval.start:
        pop pq                # earliest room freed up -> reuse it
    push interval.end         # occupy a room until this meeting ends
return pq size               # rooms in use simultaneously
```

```
class Solution {
    public int minMeetingRooms(int[][] intervals) {
        Arrays.sort(intervals, (a, b) -> a[0] - b[0]);           // sort by START
        PriorityQueue<Integer> pq = new PriorityQueue<>();         // min-heap of end times
        for (int[] interval : intervals) {
            if (!pq.isEmpty() && pq.peek() <= interval[0])       // ← reuse: earliest room is free
                pq.poll();                                         // ← occupy room until this end
            pq.offer(interval[1]);
        }
        return pq.size();                                         // ← rooms in use = heap size
    }
}
```

Time $O(n \log n)$ | **Space** $O(n)$

Sort-by-End vs Sort-by-Start+Heap

	Sort by End	Sort by Start + Heap
Goal:	count non-overlapping	count simultaneous active
Tracks:	lastEnd (scalar)	all active end times (heap)
Question answered:	max intervals I can keep	min rooms (resources) needed
Examples:	#435, #452, #646	#253 Meeting Rooms

Part 4 — Non-interval Greedy

#55 Jump Game

Description: Each element is the max jump from that position. Can you reach the last index?

Intuition: Sweep left to right tracking the farthest index reachable; if you ever stand past that frontier you're stuck.

Key: track the farthest index reachable so far. If current index exceeds it, you're stuck.

Variables: `nums` = max jump length from each index · `maxReach` = farthest index reachable so far **Pseudocode:**

```

maxReach = 0
for i from 0 to n-1:
    if i > maxReach: # standing past the frontier -> stuck
        return false
    maxReach = max(maxReach, i + nums[i])
return true

```

```

class Solution {
    public boolean canJump(int[] nums) {
        int maxReach = 0;
        for (int i = 0; i < nums.length; i++) {
            if (i > maxReach) return false; // ← can't reach index i
            maxReach = Math.max(maxReach, i + nums[i]);
        }
        return true;
    }
}

```

Time $O(n)$ | **Space** $O(1)$

#45 Jump Game II

Description: Minimum jumps to reach the last index.

Intuition: Treat each jump as a BFS layer — extend through the whole current reach before incrementing the jump count.

Key: BFS layer-by-layer. `currentEnd` = farthest we can reach with current jumps. `farthest` = farthest reachable with one more jump from anywhere in current layer. When we reach `currentEnd`, we must jump.

Variables: `nums` = max jump length per index · `jumps` = jump count · `currentEnd` = end of current BFS layer · `farthest` = farthest reachable from this layer **Pseudocode:**

```

jumps = 0, currentEnd = 0, farthest = 0
for i from 0 to n-2:
    farthest = max(farthest, i + nums[i])
    if i == currentEnd: # reached end of current layer
        jumps++
        currentEnd = farthest # extend to next layer
return jumps

```

```

class Solution {
    public int jump(int[] nums) {
        int jumps = 0, currentEnd = 0, farthest = 0;
        for (int i = 0; i < nums.length - 1; i++) {
            farthest = Math.max(farthest, i + nums[i]);
            if (i == currentEnd) { // ← exhausted current layer
                jumps++;
                currentEnd = farthest; // ← extend to next layer
            }
        }
        return jumps;
    }
}

```

Time $O(n)$ | **Space** $O(1)$

#763 Partition Labels

Description: Partition string into as many parts as possible so each letter appears in at most one part. Return the size of each part.

Intuition: A partition can't close until you pass the last occurrence of every letter seen inside it, so extend `end` to that frontier and cut when you reach it.

Key: precompute the last occurrence of each character. Extend the current partition's end to `max(end, last[c])` for each character. Close partition when `i == end`.

Variables: `s` = input string · `last[c]` = last index where char `c` appears · `start` = current partition start · `end` = farthest last-occurrence seen in current partition **Pseudocode:**

```
record last[c] = last index of each char in s
start = 0, end = 0
for i from 0 to n-1:
    end = max(end, last[s[i]]) # partition must include this char's last occurrence
    if i == end:               # reached partition boundary
        add (end - start + 1) to result
        start = i + 1
return result
```

```
class Solution {
    public List<Integer> partitionLabels(String s) {
        int[] last = new int[26];
        for (int i = 0; i < s.length(); i++) last[s.charAt(i) - 'a'] = i; // ← last occurrence
        List<Integer> result = new ArrayList<>();
        int start = 0, end = 0;
        for (int i = 0; i < s.length(); i++) {
            end = Math.max(end, last[s.charAt(i) - 'a']); // ← must include last of this char
            if (i == end) { // ← reached end of partition
                result.add(end - start + 1);
                start = i + 1;
            }
        }
        return result;
    }
}
```

Time $O(n)$ | **Space** $O(1)$

#406 Queue Reconstruction by Height

Description: People given as `[h, k]` (height `h`, `k` taller/equal people in front). Reconstruct the queue.

Intuition: Place tallest first so that inserting a person at index `k` is always correct — only taller/equal people are already placed, and shorter insertions later don't disturb their counts.

Key: sort descending by height (ties: ascending by `k`). Insert each person at index `k`. Since taller people are placed first, inserting at position `k` is always valid — there are already exactly `k` people $\geq$ this height already placed.

Variables: `people` = `[height, k]` pairs · `result` = queue being built by insertion at index `k` **Pseudocode:**

```
sort people: height DESC, ties by k ASC
result = empty list
for each person p (tallest first):
    insert p at index p.k # only taller/equal people precede it
return result
```

```

class Solution {
    public int[][] reconstructQueue(int[][] people) {
        Arrays.sort(people, (a, b) -> a[0] != b[0] ? b[0] - a[0] : a[1] - b[1]); // ← tall first, ties
        List<int[]> result = new ArrayList<>();
        for (int[] p : people)
            result.add(p[1], p); // ← insert at index k
        return result.toArray(new int[0][]);
    }
}

```

Time O(n²) | Space O(n)

Greedy Pattern Identifier

Signal	Template
"minimum removal to make non-overlapping"	Sort by end, keep non-overlapping (Template 1)
"minimum resources (arrows, groups)"	Sort by end, count groups (Template 1)
"merge overlapping intervals"	Sort by start, extend or add (Template 2)
"minimum rooms/machines"	Sort by start + min-heap (Template 3)
"can you reach the end?"	Track max reachable index
"minimum steps to reach the end?"	BFS layers via <code>currentEnd</code>
"partition string by last occurrence"	Track <code>last[]</code> array + extend <code>end</code>
"reconstruct order by two constraints"	Sort by one constraint, insert by the other
"can you complete the circuit / circular route?"	Track cumulative tank; reset start when tank < 0
"can person attend all meetings (no overlap)?"	Sort by start, compare adjacent intervals

Part 5 — Overlap & Circular Patterns

#252 Meeting Rooms

Description: Given an array of meeting intervals, determine if a person could attend all meetings (no two meetings overlap).

Intuition: After sorting by start, any overlap must be between adjacent intervals, so one linear scan comparing neighbors is enough.

Algorithm: Sort by start time. If any meeting starts before the previous one ends, there is an overlap — return false.

Variables: `intervals` = meeting `[start, end]` pairs **Pseudocode:**

```

sort intervals by START
for i from 1 to n-1:
    if intervals[i].start < intervals[i-1].end:    # overlaps previous
        return false
return true

```



```

class Solution {
    public boolean canAttendMeetings(int[][] intervals) {
        Arrays.sort(intervals, (a, b) -> a[0] - b[0]); // ← sort by start time
        for (int i = 1; i < intervals.length; i++)
            if (intervals[i][0] < intervals[i-1][1]) return false; // ← VARIATION: start < previous end
        return true;
    }
}

```

Time $O(n \log n)$ | **Space** $O(1)$

#134 Gas Station

Description: There are n gas stations on a circular route. `gas[i]` is the gas at station i ; `cost[i]` is the gas needed to travel from i to $i+1$. Return the starting station index if you can complete the circuit, or -1 if not.

Intuition: If the tank ever drops below 0, no station between the old start and here can work, so jump the start past the failure point; a valid start exists iff total gas $\geq$ total cost.

Algorithm: Greedy. Track cumulative tank. When tank goes negative, reset start to $i+1$ and reset tank. If total gas $\geq$ total cost, a valid start exists (the last reset start).

Variables: `gas / cost` = gas at / cost to leave each station · `total` = net gas over whole loop · `tank` = running gas since current `start` · `start` = candidate starting station **Pseudocode:**

```

total = 0, tank = 0, start = 0
for i from 0 to n-1:
    diff = gas[i] - cost[i]
    total += diff
    tank += diff
    if tank < 0:           # cannot reach i+1 from current start
        start = i + 1     # next station becomes the candidate
        tank = 0
return start if total >= 0 else -1

```

```

class Solution {
    public int canCompleteCircuit(int[] gas, int[] cost) {
        int total = 0, tank = 0, start = 0;
        for (int i = 0; i < gas.length; i++) {
            int diff = gas[i] - cost[i];
            total += diff;
            tank += diff;
            if (tank < 0) {
                start = i + 1; // ← VARIATION: reset start when tank goes negative
                tank = 0;
            }
        }
        return total >= 0 ? start : -1; // ← VARIATION: if total >= 0, solution exists
    }
}

```

Time $O(n)$ | **Space** $O(1)$

Additional Reference Problems

#135 Candy

Description: Assign minimum candies to children in a line: each child gets at least 1 candy, and a child with a higher rating than an adjacent neighbor gets more candies than that neighbor.

Intuition: Two passes capture the two directions of the constraint — a left-to-right pass enforces "higher than left neighbor gets more", a right-to-left pass enforces "higher than right neighbor gets more". Taking the max of both satisfies both.

Algorithm: Initialize all candies to 1. Sweep left→right giving `candies[i] = candies[i-1] + 1` when `ratings[i] > ratings[i-1]`. Sweep right→left taking `max(candies[i], candies[i+1] + 1)` when `ratings[i] > ratings[i+1]`. Sum.

Variables: `ratings` = child ratings · `candies[i]` = candies given to child `i` · `result` = total candies **Pseudocode:**

```
candies = all 1s
left-to-right, i from 1:
    if ratings[i] > ratings[i-1]:
        candies[i] = candies[i-1] + 1      # higher than left neighbor
right-to-left, i from n-2:
    if ratings[i] > ratings[i+1]:
        candies[i] = max(candies[i], candies[i+1] + 1)  # higher than right neighbor
result = sum of candies
return result
```

```
class Solution {
    public int candy(int[] ratings) {
        int n = ratings.length;
        int[] candies = new int[n];
        Arrays.fill(candies, 1);
        for (int i = 1; i < n; i++) {
            if (ratings[i] > ratings[i - 1]) {                // ← VARIATION: higher than left neighb
                candies[i] = candies[i - 1] + 1;
            }
        }
        for (int i = n - 2; i >= 0; i--) {
            if (ratings[i] > ratings[i + 1]) {                // ← VARIATION: higher than right neigh
                candies[i] = Math.max(candies[i], candies[i + 1] + 1);
            }
        }
        int result = 0;
        for (int i = 0; i < n; i++) {
            result += candies[i];
        }
        return result;
    }
}
```

Time $O(n)$ | **Space** $O(n)$

#274 H-Index

Description: Given an array of citation counts, find the largest h such that at least h papers each have at least h citations.

Intuition: After sorting ascending, if a paper at index i has $\text{citations}[i]$ citations, then there are $n - i$ papers with at least that many. The h -index is found where $\text{citations}[i] \geq n - i$ first holds.

Algorithm: Sort ascending. Scan; the first index i where $\text{citations}[i] \geq n - i$ gives $h = n - i$, the largest valid h . If none, h is 0.

Variables: citations = citation counts · n = number of papers · $n - i$ = papers with at least $\text{citations}[i]$ citations

Pseudocode:

```
sort citations ascending
for i from 0 to n-1:
    if citations[i] >= n - i:    # n-i papers each have >= this many citations
        return n - i
return 0
```

```
class Solution {
    public int hIndex(int[] citations) {
        Arrays.sort(citations);                // ← sort ascending
        int n = citations.length;
        for (int i = 0; i < n; i++) {
            if (citations[i] >= n - i) {        // ← VARIATION: n-i papers have >= cita
                return n - i;
            }
        }
        return 0;
    }
}
```

Time $O(n \log n)$ | **Space** $O(1)$

#455 Assign Cookies

Description: Each child i has a greed factor $g[i]$ (minimum cookie size needed); each cookie j has size $s[j]$. Each cookie satisfies at most one child. Maximize the number of content children.

Intuition: Sort both arrays; give the smallest sufficient cookie to the least greedy child. This never wastes a large cookie on a child a small one could satisfy.

Algorithm: Sort g and s . Two pointers: advance the cookie pointer always; advance the child pointer (counting a satisfied child) only when the current cookie satisfies the current child.

Variables: g = children greed factors · s = cookie sizes · i = child pointer · j = cookie pointer · result = content children count **Pseudocode:**

```
sort g ascending; sort s ascending
i = 0, j = 0, result = 0
while i < g.length and j < s.length:
    if s[j] >= g[i]:    # cookie j satisfies child i
        result++
        i++            # move to next child
    j++                # always move to next cookie
return result
```

```

class Solution {
    public int findContentChildren(int[] g, int[] s) {
        Arrays.sort(g);
        Arrays.sort(s);
        int i = 0, j = 0, result = 0;
        while (i < g.length && j < s.length) {
            if (s[j] >= g[i]) {
                result++;
                i++;
            }
            j++;
        }
        return result;
    }
}

```

// ← sort children by greed
// ← sort cookies by size
// ← cookie satisfies child

Time $O(n \log n)$ | **Space** $O(1)$

#575 Distribute Candies

Description: Distribute n candies (the holder eats exactly $n/2$) to maximize the number of distinct candy types eaten.

Intuition: You can eat at most $n/2$ candies, and at best one of each distinct type. So the answer is the smaller of the number of distinct types and $n/2$.

Algorithm: Count distinct types with a set. Return $\min(\text{distinct types}, n/2)$.

Variables: `candyType` = candy type per candy · `types` = set of distinct types · `candyType.length / 2` = max candies the holder may eat **Pseudocode:**

```

types = set()
for each candy: add its type to types
return min(number of distinct types, total / 2)

```

```

class Solution {
    public int distributeCandies(int[] candyType) {
        Set<Integer> types = new HashSet<>();
        for (int i = 0; i < candyType.length; i++) {
            types.add(candyType[i]);
        }
        return Math.min(types.size(), candyType.length / 2);
    }
}

```

// ← VARIATION: capped by half the total

Time $O(n)$ | **Space** $O(n)$

#605 Can Place Flowers

Description: Given a flowerbed (array of 0s and 1s where 1 is a planted flower), determine if n new flowers can be planted without any two flowers being adjacent.

Intuition: Greedily plant a flower at the first empty plot whose neighbors are also empty — planting as early as possible never blocks a future placement that a later choice would allow.

Algorithm: Scan; at each empty plot whose left and right (treating out-of-bounds as empty) are empty, plant it (set to 1) and decrement n . Succeed early if $n \leq 0$.

Variables: `flowerbed` = 0/1 plots · `n` = flowers still to plant · `leftEmpty` / `rightEmpty` = whether neighbors are empty (out-of-bounds counts as empty) **Pseudocode:**

```
for i from 0 to length-1:
    leftEmpty = (i == 0) or flowerbed[i-1] == 0
    rightEmpty = (i == last) or flowerbed[i+1] == 0
    if flowerbed[i] == 0 and leftEmpty and rightEmpty:
        plant: flowerbed[i] = 1
        n--
    if n <= 0: return true
return n <= 0
```

```
class Solution {
    public boolean canPlaceFlowers(int[] flowerbed, int n) {
        for (int i = 0; i < flowerbed.length; i++) {
            boolean leftEmpty = (i == 0) || (flowerbed[i - 1] == 0);
            boolean rightEmpty = (i == flowerbed.length - 1) || (flowerbed[i + 1] == 0);
            if (flowerbed[i] == 0 && leftEmpty && rightEmpty) { // ← VARIATION: plot and both neighbors
                flowerbed[i] = 1;                                // ← plant greedily
                n--;
            }
            if (n <= 0) {
                return true;
            }
        }
        return n <= 0;
    }
}
```

Time $O(n)$ | **Space** $O(1)$

#630 Course Schedule III

Description: Each course has `(duration, lastDay)`. Starting at day 0 and taking courses one at a time, find the maximum number of courses that can be taken so each finishes by its `lastDay`.

Intuition: Sort by deadline so deadlines are considered in order. Greedily take each course; if total time exceeds the current deadline, drop the previously taken course with the largest duration (a max-heap) — swapping out the longest course frees the most time while keeping the count.

Algorithm: Sort by `lastDay`. Maintain a max-heap of taken durations and running `time`. Add each course; if `time > lastDay`, pop the longest course and subtract it. Heap size is the answer.

Variables: `courses` = `[duration, lastDay]` · `maxHeap` = durations of taken courses (largest on top) · `time` = total time spent so far **Pseudocode:**

```
sort courses by lastDay (deadline)
maxHeap = empty, time = 0
for each course [duration, lastDay]:
    time += duration
    push duration onto maxHeap
    if time > lastDay:          # overshoot deadline
        time -= pop longest duration # drop the longest taken course
return maxHeap size
```

```

class Solution {
    public int scheduleCourse(int[][] courses) {
        Arrays.sort(courses, (a, b) -> a[1] - b[1]);           // ← sort by deadline
        PriorityQueue<Integer> maxHeap = new PriorityQueue<>((a, b) -> b - a);
        int time = 0;
        for (int[] course : courses) {
            time += course[0];
            maxHeap.offer(course[0]);
            if (time > course[1]) {                             // ← VARIATION: over deadline → drop lo
                time -= maxHeap.poll();
            }
        }
        return maxHeap.size();
    }
}

```

Time $O(n \log n)$ | **Space** $O(n)$

#881 Boats to Save People

Description: Each boat carries at most 2 people with total weight at most `limit`. Given people's weights, find the minimum number of boats to rescue everyone.

Intuition: Pair the heaviest remaining person with the lightest remaining one if they fit together; otherwise the heaviest goes alone. Sorting plus two pointers makes this pairing optimal.

Algorithm: Sort ascending. Two pointers from both ends; if `people[i] + people[j] <= limit` the lightest also boards (advance `i`). Always send the heaviest (decrement `j`) and count a boat.

Variables: `people` = weights · `limit` = boat weight cap · `i` = lightest pointer · `j` = heaviest pointer · `result` = boat count
Pseudocode:

```

sort people ascending
i = 0, j = n-1, result = 0
while i <= j:
    if people[i] + people[j] <= limit:
        i++           # lightest shares boat with heaviest
    j--               # heaviest always boards
    result++          # one boat used
return result

```

```

class Solution {
    public int numRescueBoats(int[] people, int limit) {
        Arrays.sort(people);           // ← sort ascending
        int i = 0, j = people.length - 1, result = 0;
        while (i <= j) {
            if (people[i] + people[j] <= limit) {           // ← lightest fits with heaviest
                i++;
            }
            j--;                                             // ← heaviest always boards
            result++;
        }
        return result;
    }
}

```

Time $O(n \log n)$ | **Space** $O(1)$

#986 Interval List Intersections

Description: Given two lists of pairwise-disjoint, sorted intervals, return the intersection of the two lists.

Intuition: Two pointers walk both sorted lists together. The overlap of the two current intervals is $[\max(\text{starts}), \min(\text{ends})]$, which is valid only when $\max(\text{starts}) \leq \min(\text{ends})$. Advance whichever interval ends first.

Algorithm: Two pointers i, j . Compute $lo = \max(\text{starts})$, $hi = \min(\text{ends})$; if $lo \leq hi$, record $[lo, hi]$. Advance the pointer of the interval with the smaller end.

Variables: $\text{firstList} / \text{secondList}$ = two sorted disjoint interval lists · i / j = pointers into each list · lo = latest start · hi = earliest end **Pseudocode:**

```
i = 0, j = 0
while i < firstList.length and j < secondList.length:
    lo = max(firstList[i].start, secondList[j].start)
    hi = min(firstList[i].end, secondList[j].end)
    if lo <= hi: record [lo, hi]      # valid overlap
    if firstList[i].end < secondList[j].end: i++  # advance the one ending first
    else: j++
return result
```

```
class Solution {
    public int[][] intervalIntersection(int[][] firstList, int[][] secondList) {
        List<int[]> result = new ArrayList<>();
        int i = 0, j = 0;
        while (i < firstList.length && j < secondList.length) {
            int lo = Math.max(firstList[i][0], secondList[j][0]); // ← latest start
            int hi = Math.min(firstList[i][1], secondList[j][1]); // ← earliest end
            if (lo <= hi) {                                         // ← VARIATION: valid overlap
                result.add(new int[]{lo, hi});
            }
            if (firstList[i][1] < secondList[j][1]) {              // ← advance interval that ends first
                i++;
            } else {
                j++;
            }
        }
        return result.toArray(new int[0][]);
    }
}
```

Time $O(m + n)$ | **Space** $O(m + n)$

#1005 Maximize Sum Of Array After K Negations

Description: Given an array and k , perform exactly k operations, each negating one element (the same index may be chosen repeatedly), to maximize the array sum.

Intuition: Flipping the most negative number first yields the biggest gain. After all negatives are flipped (or k runs out), any leftover flips should land on the smallest-magnitude element, since flipping it twice is a no-op.

Algorithm: Sort ascending. Flip negatives left to right while $k > 0$. If k is still positive and odd, negate the now-smallest absolute value element once. Sum.

Variables: nums = array · k = flips remaining · sum = running total · min = smallest element after flipping negatives **Pseudocode:**

```

sort nums ascending
for i while i < n and k > 0:
    if nums[i] < 0:          # flip most-negative numbers first
        nums[i] = -nums[i]
        k--
sum = sum of nums; min = smallest element
if k is odd (leftover flips):
    sum -= 2 * min          # one extra flip lands on smallest magnitude
return sum

```

```

class Solution {
    public int largestSumAfterKNegations(int[] nums, int k) {
        Arrays.sort(nums);                                // ← sort ascending
        for (int i = 0; i < nums.length && k > 0; i++) {
            if (nums[i] < 0) {                             // ← flip most-negative first
                nums[i] = -nums[i];
                k--;
            }
        }
        int sum = 0, min = Integer.MAX_VALUE;
        for (int i = 0; i < nums.length; i++) {
            sum += nums[i];
            min = Math.min(min, nums[i]);
        }
        if (k % 2 == 1) {                                  // ← VARIATION: leftover odd flips hit
            sum -= 2 * min;
        }
        return sum;
    }
}

```

Time $O(n \log n)$ | **Space** $O(1)$

#1094 Car Pooling

Description: Given trips `[numPassengers, from, to]` and a car capacity, determine whether all trips can be completed without exceeding capacity at any point.

Intuition: A difference array (sweep line) over locations records passengers boarding at `from` and leaving at `to`. Running the prefix sum gives the occupancy at each point; if it ever exceeds capacity, fail.

Algorithm: Add `num` at `from` and subtract `num` at `to` in a difference array. Prefix-sum across locations; return false if any running total exceeds capacity.

Variables: `trips = [numPassengers, from, to]` · `capacity = car cap` · `diff` = difference array over locations · `passengers` = running occupancy **Pseudocode:**

```

diff = array over locations, all 0
for each trip [num, from, to]:
    diff[from] += num    # board
    diff[to]   -= num    # leave
passengers = 0
for each location i in order:
    passengers += diff[i]    # prefix sum = current occupancy
    if passengers > capacity: return false
return true

```



```

class Solution {
    public boolean carPooling(int[][] trips, int capacity) {
        int[] diff = new int[1001];
        for (int[] trip : trips) {
            diff[trip[1]] += trip[0];           // ← board at from
            diff[trip[2]] -= trip[0];           // ← leave at to
        }
        int passengers = 0;
        for (int i = 0; i < diff.length; i++) {
            passengers += diff[i];               // ← running occupancy
            if (passengers > capacity) {         // ← VARIATION: over capacity → fail
                return false;
            }
        }
        return true;
    }
}

```

Time $O(n + \text{maxLocation})$ | **Space** $O(\text{maxLocation})$

#1272 Remove Interval

Description: Given a sorted list of disjoint intervals and an interval `toBeRemoved`, return the set of intervals after removing every point that lies inside `toBeRemoved`.

Intuition: Each interval is either fully outside the removal range (keep as is) or partially overlaps it (keep the non-overlapping left and/or right slivers). Walk once and emit the surviving pieces.

Algorithm: For each interval, if it is entirely before or after `toBeRemoved`, keep it. Otherwise keep the left piece `[start, toBeRemoved[0]]` if it has positive length and the right piece `[toBeRemoved[1], end]` if it has positive length.

Variables: `intervals` = sorted disjoint intervals · `toBeRemoved` = `[lo, hi]` range to cut out · `result` = surviving pieces
Pseudocode:

```

for each interval:
    if interval.end <= toBeRemoved.lo or interval.start >= toBeRemoved.hi:
        keep whole interval    # no overlap
    else:
        if interval.start < toBeRemoved.lo:
            keep left sliver [interval.start, toBeRemoved.lo]
        if interval.end > toBeRemoved.hi:
            keep right sliver [toBeRemoved.hi, interval.end]
return result

```

```

class Solution {
    public List<List<Integer>> removeInterval(int[][] intervals, int[] toBeRemoved) {
        List<List<Integer>> result = new ArrayList<>();
        for (int[] interval : intervals) {
            if (interval[1] <= toBeRemoved[0] || interval[0] >= toBeRemoved[1]) {
                result.add(Arrays.asList(interval[0], interval[1])); // ← no overlap → keep whole
            } else {
                if (interval[0] < toBeRemoved[0]) { // ← VARIATION: keep left sliver
                    result.add(Arrays.asList(interval[0], toBeRemoved[0]));
                }
                if (interval[1] > toBeRemoved[1]) { // ← VARIATION: keep right sliver
                    result.add(Arrays.asList(toBeRemoved[1], interval[1]));
                }
            }
        }
        return result;
    }
}

```

Time $O(n)$ | **Space** $O(n)$

#1326 Minimum Number of Taps to Open to Water a Garden

Description: A garden spans $[0, n]$. Tap i at position i waters $[i - \text{ranges}[i], i + \text{ranges}[i]]$. Find the minimum number of taps to water the whole garden, or -1 if impossible.

Intuition: This is a jump-game in disguise: for each starting point record the farthest reach of any tap covering it, then greedily extend coverage layer by layer like minimum jumps.

Algorithm: Build $\text{maxReach}[i]$ = farthest right coverage of any tap whose left edge is at or before i . Sweep like Jump Game II: track currentEnd and farthest ; when reaching currentEnd , increment taps and jump. Fail if farthest stalls before n .

Variables: n = garden length · ranges = tap reach radius per position · $\text{maxReach}[\text{left}]$ = farthest right covered starting at left · taps = taps opened · currentEnd / farthest = jump-game layer bounds **Pseudocode:**

```

for each position i:
    left = max(0, i - ranges[i]); right = min(n, i + ranges[i])
    maxReach[left] = max(maxReach[left], right)
taps = 0, currentEnd = 0, farthest = 0
for i from 0 to n:
    if i > farthest: return -1      # gap in coverage
    farthest = max(farthest, maxReach[i])
    if i == currentEnd and i < n:  # end of current layer
        taps++
        currentEnd = farthest
return taps if currentEnd >= n else -1

```

```

class Solution {
    public int minTaps(int n, int[] ranges) {
        int[] maxReach = new int[n + 1];
        for (int i = 0; i <= n; i++) {
            int left = Math.max(0, i - ranges[i]);
            int right = Math.min(n, i + ranges[i]);
            maxReach[left] = Math.max(maxReach[left], right); // ← farthest reach from this left edge
        }
        int taps = 0, currentEnd = 0, farthest = 0;
        for (int i = 0; i <= n; i++) {
            if (i > farthest) { // ← VARIATION: gap in coverage → impos
                return -1;
            }
            farthest = Math.max(farthest, maxReach[i]);
            if (i == currentEnd && i < n) { // ← exhausted current layer → open a t
                taps++;
                currentEnd = farthest;
            }
        }
        return currentEnd >= n ? taps : -1;
    }
}

```

Time $O(n)$ | **Space** $O(n)$

#1353 Maximum Number of Events That Can Be Attended

Description: Each event $[start, end]$ can be attended on any single day in $[start, end]$. Attend at most one event per day. Find the maximum number of events you can attend.

Intuition: Process days in increasing order; among all events available on the current day, attend the one that ends soonest (a min-heap by end day) — saving longer-lived events for later days.

Algorithm: Sort events by start. For each day, push all events that have started into a min-heap keyed on end day, discard events whose end day has passed, then attend (poll) one event and count it.

Variables: `events` = $[start, end]$ · `minHeap` = end days of events available now (earliest on top) · `i` = next event index · `day` = current day · `result` = events attended **Pseudocode:**

```

sort events by START
minHeap = empty (end days), i = 0, day = 0, result = 0
while i < n or minHeap nonempty:
    if minHeap empty: day = events[i].start    # jump forward to next event
    while i < n and events[i].start <= day:    # add all events started by today
        push events[i].end; i++
    poll minHeap; result++; day++              # attend earliest-ending event
    while minHeap nonempty and top < day:      # drop expired events
        poll minHeap
return result

```

```

class Solution {
    public int maxEvents(int[][] events) {
        Arrays.sort(events, (a, b) -> a[0] - b[0]); // ← sort by start day
        PriorityQueue<Integer> minHeap = new PriorityQueue<>(); // ← min-heap of end days
        int i = 0, day = 0, result = 0, n = events.length;
        while (i < n || !minHeap.isEmpty()) {
            if (minHeap.isEmpty()) {
                day = events[i][0]; // ← jump to next event's start
            }
            while (i < n && events[i][0] <= day) { // ← add all events started by today
                minHeap.offer(events[i][1]);
                i++;
            }
            minHeap.poll(); // ← VARIATION: attend earliest-ending
            result++;
            day++;
            while (!minHeap.isEmpty() && minHeap.peek() < day) { // ← drop events that already expired
                minHeap.poll();
            }
        }
        return result;
    }
}

```

Time $O(n \log n)$ | **Space** $O(n)$

#1488 Avoid Flood in The City

Description: `rains[i] > 0` means lake `rains[i]` fills with rain on day `i`; `rains[i] == 0` is a dry day on which you may dry exactly one lake. A full lake that rains again floods. Return an array where dry days hold the lake to dry (any valid choice) and rain days hold -1, or an empty array if flooding is unavoidable.

Intuition: When a lake that is already full rains again, you must have used a dry day between its two rains. Greedily assign the earliest available dry day that falls after the lake was last filled — a `TreeSet` of dry-day indices gives that earliest day.

Algorithm: Track each full lake's last-rain day in a map and dry-day indices in a `TreeSet`. On a rain day for an already-full lake, find the smallest dry day after its last rain; if none exists, flooding is unavoidable. Use that dry day for this lake. Leftover dry days dry any lake (default 1).

Variables: `rains` = per-day rain (lake id or 0=dry) · `result` = answer array · `fullDay` = lake -> day it was last filled · `dryDays` = `TreeSet` of available dry-day indices **Pseudocode:**

```

for i from 0 to n-1:
    if rains[i] == 0:      # dry day
        add i to dryDays; result[i] = 1 (default, may overwrite)
    else:
        result[i] = -1; lake = rains[i]
        if lake already full:
            dry = smallest dry day > lake's last full day
            if no such dry day: return empty array # flood
            result[dry] = lake; remove dry from dryDays
        fullDay[lake] = i
return result

```

```

class Solution {
    public int[] avoidFlood(int[] rains) {
        int n = rains.length;
        int[] result = new int[n];
        Map<Integer, Integer> fullDay = new HashMap<>(); // ← lake → day it was last filled
        TreeSet<Integer> dryDays = new TreeSet<>(); // ← indices of available dry days
        for (int i = 0; i < n; i++) {
            if (rains[i] == 0) {
                dryDays.add(i);
                result[i] = 1; // ← default; may be overwritten
            } else {
                result[i] = -1;
                int lake = rains[i];
                if (fullDay.containsKey(lake)) { // ← lake already full → must dry it ea
                    Integer dry = dryDays.higher(fullDay.get(lake));
                    if (dry == null) { // ← VARIATION: no dry day available →
                        return new int[0];
                    }
                    result[dry] = lake; // ← use that dry day for this lake
                    dryDays.remove(dry);
                }
                fullDay.put(lake, i);
            }
        }
        return result;
    }
}

```

Time $O(n \log n)$ | **Space** $O(n)$